

Multiple Sensor Magnetic Recording Dependence on Channel Bit Density, Overhang and Reread/Reader Count for Conventional and Shingled Heat Assisted Magnetic Recording

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Multiple Sensor Magnetic Recording (MSMR) gain is dependent on channel bit density (CBD), overhang and the number of re-reads/readers. As CBD increases for the HAMR system, the MSMR gain increases. If HAMR readers are unable to scale with trackpitch, overhang will increase and the MSMR gain will decrease. The dependence of CBD and overhang on MSMR gain will compete with each other for the total MSMR gain of the HAMR system. MSMR gain will also depend on the number of re-reads and/or readers. At spinstand, we have demonstrated MSMR with 10 readers/re-reads (MSMR 10R) gain of 11% ADC with HAMR SMR with the improvement taken in trackpitch. With HAMR SMR and TDMR, we have achieved 4.5 Tfpsi areal density demonstration, approaching 8.0 TB/disk on the spinstand.

Index Terms— Heat Assisted Magnetic Recording (HAMR), Conventional Magnetic Recording (CMR), Shingled Magnetic Recording (SMR), Multiple Sensor Magnetic Recording (MSMR), Two-Dimensional Magnetic Recording (TDMR), Areal Density Capacity (ADC)

I. INTRODUCTION

HHEAT ASSISTED MAGNETIC RECORDING (HAMR) has proven to be the primary successor to perpendicular magnetic recording providing continued and significant growth in areal density [1]. The read-back architecture Multiple Sensor Magnetic Recording (MSMR) can be combined with Conventional Magnetic Recording (CMR) and Shingled Magnetic Recording (SMR) write architectures to increase areal density [2-5] by using two or more readers and/or multiple spins to read-back the same track. Previously, the MSMR 2R and 3R gain was taken in linear density and trackpitch for HAMR CMR and SMR [5].

As the HAMR reader width scales with trackpitch and Reader SNR maintained, the PW50 will decrease with an increase in the downtrack reader resolution. The Channel Bit Density (CBD) is dependent on the PW50, the linear velocity and data rate of the recording system. As the PW50 decreases and data rate increases, the CBD increases. If the reader width cannot scale with maintaining the reader SNR, overhang will increase which is defined as the reader width over the trackpitch. For this study, MT10 is used as proxy for reader width [6] with $\text{Overhang} = \text{MT10}/\text{Trackpitch}$. As the overhang increases, the adjacent track interference increases since adjacent tracks for high areal densities are very close to each other. One strategy to improve the total SNR of the recording system for narrow readers is adding more readers or re-reads of the track.

In this paper, we will show the HAMR MSMR dependence on CBD, overhang and the number of readers/re-reads.

II. SPINSTAND EXPERIMENTAL DETAILS

We investigated the ADC for HAMR CMR and SMR with single reader and MSMR Gain with 2-10 readers/re-reads on the spinstand with 6x adjacent track writes for CMR and 1x adjacent track write for SMR. The ADC demo criteria had additional margin for adjacent track interference with a tighter

requirement than the ASTC areal density metric [7]. For this study, ten integrated HAMR heads were measured on HAMR media. The HAMR heads and media were similar in design to those used in the previous study [5].

Spinstand measurements were at radius of 29 mm, skew 0° and 7200 rpm with linear velocity of ~22 m/s. Writer and reader clearances were maintained at 1 nm and compensated for changes in writer and laser protrusions. Channel areal density (Gfpsi) was measured and the channel code rate was similar to the previous studies [3-5]. The MSMR gain with 2-10 readers (MSMR 2R, MSMR 3R, ..., MSMR 10R) was calculated with multiple spin re-read captured waveforms from the spinstand with the native HAMR reader processed by MSMR reader software channel.

III. RESULTS

The spinstand results of the dependence of MSMR 2R and MSMR 3R linear density gain on CBD are shown in Fig. 1. As the CBD increases, the MSMR gain increases. The MSMR 3R gain had a steeper slope with CBD than MSMR 2R.

As the overhang of the reader increases, the MSMR 2R/3R gain decreases for both CMR and SMR. The MSMR gain for SMR decreases slower with overhang than with CMR.

The MSMR 2R to MSMR 10R linear density gain and trackpitch improvement for HAMR CMR and SMR are shown in Fig. 3. For the CMR tracks, the MSMR 10R gain was similar when taken in linear density and trackpitch improvement (6.8% ADC gain). For the SMR tracks, the MSMR 10R gain preferred trackpitch improvement (11.0% ADC gain) to linear density (7.7% ADC gain) to maximize ADC. As the number of readers/re-reads increases, the MSMR gain plateaus for both CMR and SMR.

Spinstand high ADC demonstrations using HAMR CMR and HAMR SMR are shown in Fig. 4. The highest HAMR ADC record of 4.5 Tfpsi with state-of-art HAMR heads and media is shown. The estimated capacity per disk for this ADC demonstration was approaching 8.0 TB/disk based on expected format efficiency. This ADC demonstration used HAMR SMR

with TDMR and demonstrates that HAMR writing and TDMR readback is possible on approximately 3.1 grains per bit compared with the ~ 3.6 grains shown previously [5]. The spinstand demonstration represents what is possible in a recording system and that there are no fundamental physic barriers to achieving over 4.5 Tfpsi or 8 TB/disk with HAMR technology with TDMR.

IV. CONCLUSION

HAMR MSMR gain was demonstrated to be dependent on CBD, overhang and the number of readers/re-reads. As CBD increases, the MSMR gain increases. If the HAMR reader widths are unable to scale with trackpitch, overhang will increase and the MSMR gain will decrease. The dependence of CBD and overhang will compete with each other for the total MSMR gain of the HAMR system. The MSMR gain will increase with increasing number of readers/re-reads and the gain will plateau. At spinstand, 4.5 Tfpsi areal density demonstration, equating to around 8.0 TB/disk, has been achieved with HAMR SMR with TDMR. These experiments help to establish the intrinsic scalability of HAMR with significant and continued growth in areal density as the number of grains drastically reduces for the recording system.

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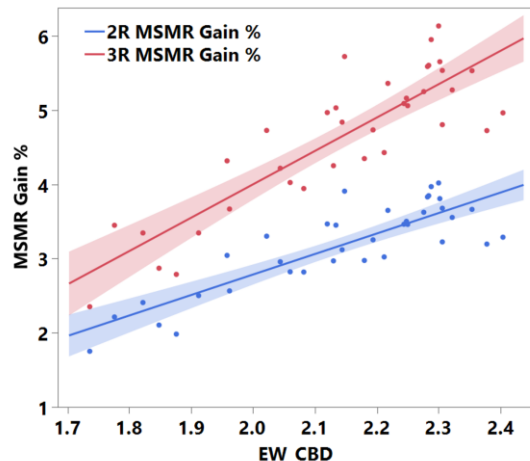


Fig. 1. Dependence of HAMR MSMR 2R/3R Gain on Channel Bit Density

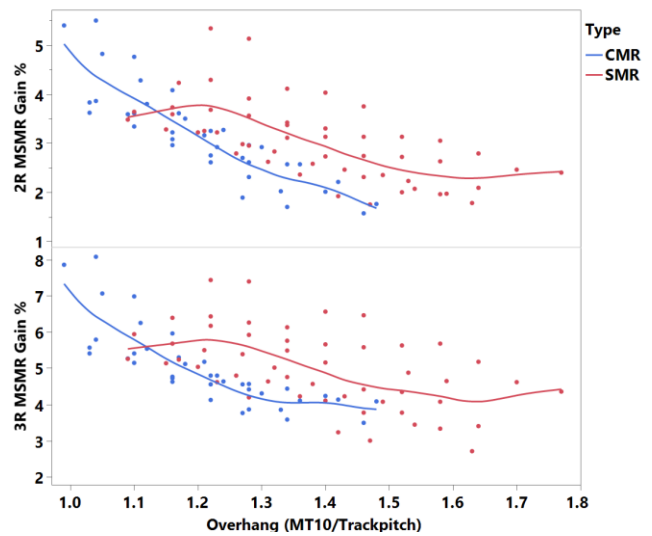


Fig. 2. Dependence of HAMR CMR & SMR with MSMR 2R/3R Gain on Reader Overhang

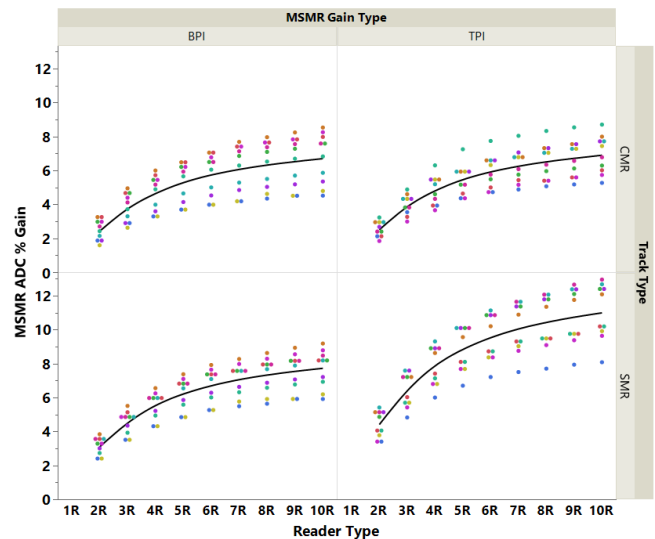


Fig. 3. HAMR CMR & SMR Tracks with MSMR 2R to 10R Gain in ADC

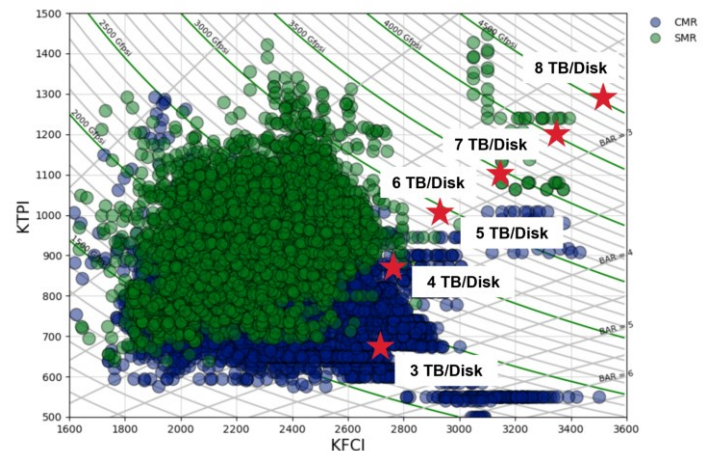


Fig. 4. HAMR Areal Density Capability Demonstrations using HAMR CMR and SMR and TDMR